

Pratham Arora

pratham3992@gmail.com | [linkedin.com/in/pratham3992arora](https://www.linkedin.com/in/pratham3992arora) | github.com/prats3992 | pratham-arora.vercel.app

EDUCATION

Plaksha University

B.Tech. in Computer Science & AI — CGPA: 8.19 / 10.0

Mohali, India

Expected Aug. 2026

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenCV, Segment Anything (SAM), GANs

Languages: Python (NumPy, Pandas), C++, MATLAB, SQL, JavaScript, TypeScript

Cloud & Tools: Azure (Cosmos DB, OpenAI), Git/GitHub, Linux, Firebase, RAG Pipelines, Vercel

Frameworks: FastAPI, Next.js, Unity (C#), Jupyter, VS Code, Google Colab

RESEARCH EXPERIENCE

Undergraduate Researcher

June 2023 – Aug. 2023

Supervised by Prof. Sandeep Manjanna | Plaksha University

- Developed preprocessing pipeline to optimize **Segment Anything Model (SAM)** for agricultural crop-weed segmentation, improving mean IoU from 0.67 to 0.75 (12% improvement) on sparse datasets ($N = 500$ images).
- Reduced computer vision pipeline runtime from 8.5 hours to 5.1 hours (40% speedup) by implementing vectorized NumPy operations and multiprocessing for batch processing 10,000+ image datasets.

INDUSTRY EXPERIENCE

AI Intern

June 2025 – July 2025

Cotality | Kolkata

- Built **FastAPI** backend processing 50,000+ lines of code daily across 15+ programming languages using Abstract Syntax Tree (AST) analysis, reducing engineering team's documentation time by an estimated 60%.
- Implemented hash-based change detection system for **vector database updates**, chunking codebases function-by-function and tracking content hashes to skip unchanged functions – reducing embedding generation costs by 85%.

SDE Intern (Robotics)

June 2024 – July 2024

Orangewood Labs | Noida

- Refactored 15,000-line monolithic **RoboGPT** codebase into modular architecture with 40% fewer inter-module dependencies, enabling parallel development across 3 teams.
- Integrated Computer Vision capabilities into robotic control loop, optimizing calibration routines and improving robotic arm positioning accuracy for complex food preparation tasks.

SELECTED PROJECTS

VR LLM Conversational Agent | Unity, Gemini 2.5 Flash, HCI, Statistical Analysis

Aug. 2025 – Dec. 2025

- Engineered real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud Speech-to-Text and Text-to-Speech, achieving average response latency of 1.8s through parallel API calls and caching.
- Designed **within-subjects experimental study** ($N = 18$) to evaluate three latency feedback mechanisms (embodied gestures, visual indicators, verbal fillers) on user perception using Godspeed questionnaire and custom metrics.
- Demonstrated through **statistical analysis (Wilcoxon signed-rank test, $p < 0.05$)** that embodied gestures significantly improved perceived animacy and immersion compared to explicit loading indicators.
- Co-authored research paper** (in preparation for submission to Presence/India HCI/TOCHI) analyzing component-level latency feedback mechanisms in embodied AI agents.

Pre-eclampsia Discourse Analysis | NLP, VADER, LDA, Data Engineering

Aug. 2025 – Dec. 2025

- Built automated data pipeline using **PRAW** to scrape 22,000+ interactions from 19 pregnancy-related subreddits, implementing concurrent scraping with rate limiting for 99.5% data completeness.
- Discovered 19.4% sentiment shift ("Buffer Effect") in pre-eclampsia discussions using **VADER sentiment analysis** and **LDA topic modeling**, identified 35.8x volume increase post-pandemic indicating growing health anxiety.

UI Wireframe Generator | Python, PyTorch, Meta LLaMA, Computer Vision

Aug. 2024 – Dec. 2024

- Explored generative approaches for UI wireframe synthesis from text prompts using **RICO dataset** (72,000+ UI screens), experimenting with Cross-Attention GAN (CAGAN) modified from SAGAN architecture.
- Fine-tuned **LLaMA-7B model** on custom text-UI pairs to translate natural language requirements into structured JSON layouts, ultimately achieving best results with LLM-based approach due to GPU constraints.

Kelp Forest Segmentation | Python, PyTorch, U-Net, Satellite Imagery

Jan. 2024 – May 2024

- Achieved **Top 100 ranking (83rd percentile)** in Kaggle satellite imagery competition using **U-Net with EfficientNet-B3** backbone for semantic segmentation of kelp forests in noisy satellite imagery.

- Engineered preprocessing pipeline using custom **CNN filters** to detect and remove streaky atmospheric artifacts from satellite images, improving model training stability.

AI Resume & Interview Coach | *Next.js, Gemini 2.5 Flash API, Prompt Engineering*

Jan. 2025 – April 2025

- Built full-stack AI resume builder serving 500+ users, integrated **Gemini 2.5 Flash API** to power "Interview GPT" generating 1,000+ personalized DSA questions with detailed solutions.
- Implemented **Firebase Authentication** with Google OAuth supporting 200+ active users, achieving 4.2/5 average rating from 80+ user submissions.

LEADERSHIP

Head of Tech

April 2025 – Present

Athleda Sports Society | Plaksha Univ.

- Lead 4-person dev team building sports platform serving 500+ students – established CI/CD pipeline reducing deployment time from 45 minutes to 8 minutes.

Head of Web Development

Feb. 2024 – Feb. 2025

LEAP.AI@Plaksha | Plaksha Univ.

- Managed 6-person web team building AI club website attracting 800+ monthly users – mentored 6 junior developers in Next.js, TypeScript, and Vercel deployment.